

Chase Mateusiak

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Dear Michigan Medicine Hiring Committee,

I am a Bioinformatics Scientist at Washington University in St. Louis with six years of experience in multi-omics analysis, machine learning, and clinical translational research. My family relocated to Ann Arbor in early 2025, and I am now looking for opportunities here. This position appears to be a good fit, and I am excited to apply.

My work spans the full preprocessing and analysis stack across RNA-seq and multiple modalities of chromatin profiling assays, applied in both basic research (transcription factor network inference) and clinical contexts (ICU patient cohorts). I've implemented workflows at scale on SLURM clusters and in the cloud with Nextflow pipelines, including a published nf-core workflow. I am also experienced building Shiny dashboards and formally packaging and distributing software via Bioconductor, PyPI, and Bioconda.

Sincerely,
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WORK

Bioinformatics Scientist

2020–present

Washington University, St. Louis

I am a Bioinformatics Scientist for two groups. I have been in the Brent lab since 2020, and joined the Sinha group in January of 2026.

- Michael Brent's lab is in the Center for Genome Sciences & Systems Biology. The Brent lab studies methods of network inference, primarily transcription factor network inference. I perform data management, automate processing workflows, and contribute to the development and distribution of novel network inference methods. My current projects are:
 - Developing a novel method of inferring transcription factor interaction by regressing transcription factor response to either over expression or knockout onto the interaction of multiple TFs. I am the first author on a publication that is current out for review for this project.
 - Developing a harmonized, public data resource of all 11 genome wide transcription factor perturbation response and binding datasets. These data are hosted for free in a computationally efficient format on the machine learning platform HuggingFace.
- Pratik Sinha's lab in the Department of Anesthesiology. The Sinha lab primarily studies clinical data gathered from the ICU. Currently, I am working on the following projects:
 - evaluating the effectiveness of computational cell decomposition (eg CIBERSORTx) for whole blood RNAseq from seriously ill patients.
 - evaluating basis reduction techniques for their utility in analyzing longitudinal RNAseq data in sepsis patient cohorts.
 - refining a method of predicting clock time from bulk RNAseq data in patients who suffer post operative delirium.

Primary responsibilities

- Implement established, recently published, and novel analysis algorithms on bulk RNA-seq, various transcription factor binding modalities (Calling Cards, ChEC-seq, ChIP-seq) and sequence analysis in both single data type and multi-omic modes
- Responsible for data processing and QC of sequencing experiment data for multiple labs, including a large \$80 million multi-site collaboration
- Provide computational support to collaborators across organisms (yeast, mouse, human)

Notable achievements

- Developed a novel method of inferring transcription factor interaction at promoters, integrating both transcription factor binding and overexpression and knockout (Mateusiak et al, currently out for review 2026)
- Collected, harmonized, databased all 11 of the major yeast transcription factor binding and perturbation response datasets (Mateusiak et al, expected publication 2026). Notably, the data is hosted using a data lake architecture on huggingface, a machine learning platform that provides free hosting. I developed both a programmatic library and shiny app that provides researchers easy access to this rich data
- Developed, published and released on nf-core a Nextflow workflow for processing Calling Cards data in both yeast and mammals
- Implemented and manage the cloud architecture for two labs, including providing access to Seqera Platform
- Developed independent collaborations resulting in two publications
- Developed the Python curriculum implementation for Algorithms for Computational Biology, a 500-level course in the Washington University CS department
- Trained and directly supervised 5 undergraduate and master's students in machine learning based methods of multi-omics data analysis

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WORK (CONTINUED)

Research Assistant

2019

University of California, San Francisco

- Processed RNA-seq data from human skin cells and tick
- Compiled transcriptome annotations for *Ixodes scapularis*
- Processed data from a deep mutational scan screen in *E. coli*

Teacher

2012–2019

San Francisco Unified School District

- Responsible for the academic progress of 115 students per year in English and math

Civil Affairs Sergeant

2011–2019

US Army Reserve

- Trained in developing local relationships, identifying achievable development projects, and coordinating allied forces activities with local government and non-profit organizations

Teacher

2008–2010

US Peace Corps, Turkmenistan

- Developed and implemented an interactive English as a Foreign Language curriculum that improved speaking and vocabulary skills across eight classrooms for over 250 students

SKILLS

- R/python (advanced)
- Nextflow (advanced)
- Rust
- extensive experience on HPC environments (SLURM, SGE)
- AWS Cloud infrastructure, including terraform
- git/github
- open source best practice packaging in R, Python and Rust with distribution on the major repositories (bioconductor, pypi, bioconda, nf-core)
- containerization in Docker and Singularity
- Structured database development and maintenance in SQL derivatives (postgresql, sqlite) and Django/Django Rest Framework for quickly creating a DB and API
- Data lake architecture and parquet file data storage

EDUCATION

- M.S. Computer Science — Washington University, St. Louis, 2022
- B.A. English Literature and Language — University of Michigan, Ann Arbor, 2008

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PUBLICATIONS

Mateusiak, C., Jia, E., Plaggenberg, J. N., Erdenebaatar, Z., Wang, Y., Shively, C., Liao, G., Mitra, R. D., & Brent, M. R. (2026). Ubiquitous functional synergy partially explains why most transcription factor binding is non-functional. *bioRxiv* (preprint). <https://doi.org/10.64898/2026.01.19.700460>

Karagiannis, T. T., **Mateusiak, C.**, Acharya, S., Jung, W. J., Liao, S., et al. (2026). Whole blood transcriptional signatures of age and survival identified in long life family and integrative longevity omics studies. *GeroScience*. <https://doi.org/10.1007/s11357-025-02090-x>

Abid, D., Brown, H., **Mateusiak, C.**, Doering, T. L., & Brent, M. R. (2025). Mapping the transcriptional regulatory network of a fungal pathogen by exploiting transcription factor perturbation. *mBio*. <https://doi.org/10.1128/mbio.02797-25>

Kang, Y. S., Jung, J., Brown, H. L., **Mateusiak, C.**, Doering, T. L., & Brent, M. R. (2025). Leveraging a new data resource to define the response of *Cryptococcus neoformans* to environmental signals. *Genetics*, 229(1), 1–29. <https://doi.org/10.1093/genetics/iyae178>

Yen, A., **Mateusiak, C.**, Sarafinowska, S., Gachechiladze, M. A., Guo, J., Chen, X., Moudgil, A., Cammack, A. J., Hoisington-Lopez, J., Crosby, M., Brent, M. R., Mitra, R. D., & Dougherty, J. D. (2023). Calling Cards: a customizable platform to longitudinally record protein-DNA interactions over time in cells and tissues. *Current Protocols*, 3(9), e883. <https://doi.org/10.1002/cpz1.883>

Ring, K., Couper, L. I., Sapiro, A. L., Yarza, F., Yang, X. F., Clay, K., **Mateusiak, C.**, Chou, S., & Swei, A. (2022). Host blood meal identity modifies vector gene expression and competency. *Molecular Ecology*, 31(9), 2698–2711. <https://doi.org/10.1111/mec.16413>

Radkov, A., Sapiro, A. L., Flores, S., Henderson, C., Saunders, H., Kim, R., Massa, S., Thompson, S., **Mateusiak, C.**, Biboy, J., Zhao, Z., Starita, L. M., Hatleberg, W. L., Vollmer, W., Russell, A. B., Simorre, J. P., Anthony-Cahill, S., Brzovic, P., Hayes, B., & Chou, S. (2022). Antibacterial potency of type VI amidase effector toxins is dependent on substrate topology and cellular context. *eLife*, 11, e79796. <https://doi.org/10.7554/eLife.79796>